

Amazon CloudWatch

Class no:26

Amazon CloudWatch

Amazon CloudWatch is a monitoring and observability service built for DevOps engineers, developers, site reliability engineers (SREs), and IT managers. It provides data and actionable insights to monitor applications, respond to system-wide performance changes, and optimize resource utilization.

1. Core Concepts & Terminology

To understand CloudWatch, you must grasp these four primary building blocks:

- **Namespaces:** A container for CloudWatch metrics. AWS services have their own namespaces (e.g., AWS/EC2), and you can create custom ones for your apps.
- **Metrics:** The fundamental concept. A metric represents a time-ordered set of data points (e.g., CPU utilization, Disk Read Bytes).
- **Dimensions:** A name/value pair that is part of the identity of a metric. You can use dimensions to filter your data (e.g., InstanceID=i-12345678).
- **Statistics:** Metric data aggregations over specified periods (e.g., Average, Minimum, Maximum, Sum, Sample Count, and Percentiles).

2. Primary Capabilities

CloudWatch is divided into several functional areas:

CloudWatch Metrics

- **Standard Metrics:** Automatically collected from AWS services (EC2, RDS, S3, etc.) at no cost.
- **Custom Metrics:** Data you send to CloudWatch from your own applications (e.g., "Active Users" or "Page Load Time").
- **High-Resolution Metrics:** Allows you to track data with a granularity of up to **1 second**.

CloudWatch Logs

- **Log Groups & Streams:** Organize logs from various sources.
- **Logs Insights:** An interactive query language that lets you search and analyze log data instantly.
- **Vended Logs:** Logs natively published by AWS services (e.g., VPC Flow Logs, Route 53 logs).

CloudWatch Alarms

- **Static Threshold Alarms:** Triggers when a metric breaches a fixed value.
- **Anomaly Detection:** Uses machine learning to analyze historical data and identifies expected behavior; alarms fire when data falls outside the "normal" band.
- **Composite Alarms:** Combines multiple alarms into one to reduce "alarm noise."

CloudWatch Dashboards

- Customizable homepages in the AWS Console to monitor your resources in a single view.
- Can include metrics, logs, and alarm statuses across different regions.

3. Key Use Cases

Use Case	How CloudWatch Helps
Infrastructure Monitoring	Track CPU, Memory, and Disk health of EC2 and on-premises servers.
Application Monitoring	Collect custom logs and metrics to find bottlenecks in code.
Resource Optimization	Use metrics to determine if instances are over-provisioned (saving money).
Auto Scaling	Trigger EC2 Auto Scaling groups based on metric thresholds (e.g., CPU > 70%).
Compliance & Security	Monitor CloudTrail logs for unauthorized API calls or configuration changes.

4. Pricing Model (At a Glance)

CloudWatch follows a **pay-as-you-go** model with a generous **Free Tier**.

- **Free Tier:** Includes 10 metrics, 5GB of log ingestion, 3 dashboards, and 10 alarms.
- **Paid Tier:**
 - **Logs:** Charged per GB ingested and stored.
 - **Metrics:** Charged per metric per month (custom metrics).
 - **Dashboards:** \$3.00 per dashboard per month.
 - **Insights:** Charged per GB of data scanned during queries.

Hands on: Creating a cloudwatch to monitor an EC2 instance and notification via Email:

Step no:1 Create an instance named: **My-cloudWatch-instance**.

OS type: **Amazon Linux**

Launch an instance [Info](#)

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags [Info](#)

Name

My-cloudwatch-instance

[Add additional tags](#)

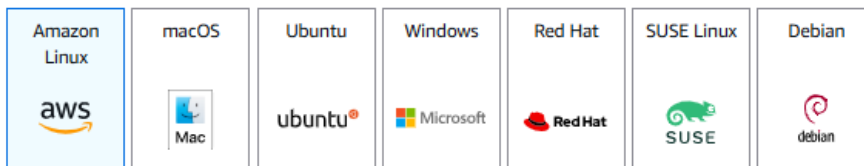
▼ Application and OS Images (Amazon Machine Image) [Info](#)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

🔍 Search our full catalog including 1000s of application and OS images

Recents

Quick Start



🔍
[Browse more AMIs](#)
Including AMIs from
AWS, Marketplace and
the Community

Amazon Machine Image (AMI)

Amazon Linux 2023 kernel-6.1 AMI
ami-0e12ffc2dd465f6e4 (64-bit (x86), uefi-preferred) / ami-0c64092ef0a77b87f (64-bit (Arm), uefi)
Virtualization: hvm ENA enabled: true Root device type: ebs

Description

Amazon Linux 2023 (kernel-6.1) is a modern, general purpose Linux-based OS that comes with 5 years of long term support. It is optimized for AWS and designed to provide a secure, stable and high-performance execution environment to develop and run your cloud applications.

Amazon Linux 2023 AMI 2023.11.20260413.0 x86_64 HVM kernel-6.1

Architecture	Boot mode	AMI ID	Publish Date	Username	i	
64-bit (x86) ▼	uefi-preferred	ami-0e12ffc2dd465f6e4	2026-04-10	ec2-user		Verified provider

Keypair: Choose an **existing keypair**

Security group: **Default security group**

Launch instance.

▼ Key pair (login) [Info](#)

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - *required*

mumkp

[Create new key pair](#)

▼ Network settings [Info](#)

[Edit](#)

Network | [Info](#)

vpc-0538c231a226fe3c9

Subnet | [Info](#)

No preference (Default subnet in any availability zone)

Auto-assign public IP | [Info](#)

Enable

Firewall (security groups) | [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☐ Create security group

☒ Select existing security group

Common security groups | [Info](#)

Select security groups

[Compare security group rules](#)

default sg-06d80e515c32b9515 X

VPC: vpc-0538c231a226fe3c9

Security groups that you add or remove here will be added to or removed from all your network interfaces.

▼ Configure storage [Info](#)

[Advanced](#)

1x 8 GiB gp3 Root volume, 3000 IOPS, Not encrypted

[Add new volume](#)

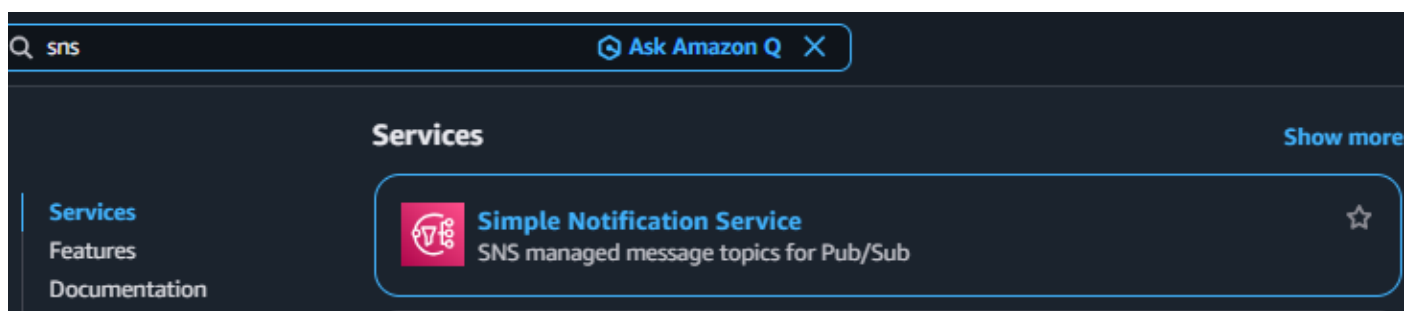
[Click refresh to view backup information](#)

The tags that you assign determine whether the instance will be backed up by any Data Lifecycle Manager policies.

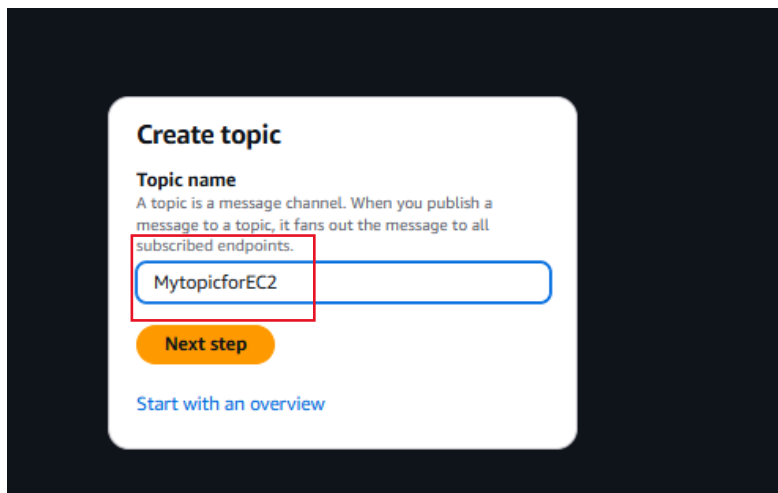
Step no:2 In the console → search: **Simple Notification Service**.

This SNS is used to send notifications whenever the resource that we are monitoring reaches the maximum threshold.

The notification can be sent as email, to our phone number, http etc.,



Once SNS is opened → Create a topic: **MytopicforEC2** → Next step.



The screenshot shows a 'Create topic' modal window. It has a title 'Create topic' and a subtitle 'Topic name'. Below the subtitle is a description: 'A topic is a message channel. When you publish a message to a topic, it fans out the message to all subscribed endpoints.' There is a text input field containing 'MytopicforEC2'. Below the input field is an orange 'Next step' button and a blue link 'Start with an overview'.

Configuration,

Type: **Standard**

Name: **MytopicforEC2**

Same name for display name

Create **Topic**.

Create topic

Details

Type | [Info](#)

Topic type cannot be modified after topic is created

☐ FIFO (first-in, first-out)

- Strictly-preserved message ordering
- Exactly-once message delivery
- Subscription protocols: SQS

☒ **Standard**

- Best-effort message ordering
- At-least once message delivery
- Subscription protocols: SQS, Lambda, Data Firehose, HTTP, SMS, email, mobile application endpoints

Name

MytopicforEC2

Maximum 256 characters. Can include alphanumeric characters, hyphens (-) and underscores (_).

Display name - *optional* | [Info](#)

To use this topic with SMS subscriptions, enter a display name. Only the first 10 characters are displayed in an SMS message.

MytopicforEC2

Maximum 100 characters.

► Tags - *optional*

A tag is a metadata label that you can assign to an Amazon SNS topic. Each tag consists of a key and an optional value. You can use tags to search and filter your topics and track your costs. [Learn more](#)

► Active tracing - *optional* | [Info](#)

Use AWS X-Ray active tracing for this topic to view its traces and service map in Amazon CloudWatch. Additional costs apply.

[Cancel](#)

[Create topic](#)

Step no:3 Once the topic has been created, we need to add subscription. Only the subscribed persons can receive email notifications.

Click: **Create subscription**

MytopicforEC2 [Edit](#) [Delete](#) [Publish message](#)

Details

Name MytopicforEC2	ARN arn:aws:sns:ap-south-1:314383175192:MytopicforEC2	Display name MytopicforEC2	Type Standard
------------------------------	---	--------------------------------------	-------------------------

Topic owner
314383175192

Subscriptions | Access policy | Data protection policy | Delivery policy (HTTP/S) | Delivery status logging | Encryption | Tags | Integrations

Subscriptions (0) [Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

Search

ID	Endpoint	Status	Protocol
No subscriptions found You don't have any subscriptions to this topic. Create subscription			

The **ARN** of the Topic will be auto populated.

Protocol: **Email**

Endpoint: **Add the respective email address** we want to send notifications.

Create Subscription.

Create subscription

Details

Topic ARN
arn:aws:sns:ap-south-1:314383175192:MytopicforEC2

Protocol
The type of endpoint to subscribe
Email

Endpoint
An email address that can receive notifications from Amazon SNS.
nithyaaisubs@gmail.com

After your subscription is created, you must confirm it. [Info](#)

► **Subscription filter policy - optional** [Info](#)
This policy filters the messages that a subscriber receives.

► **Redrive policy (dead-letter queue) - optional** [Info](#)
Send undeliverable messages to a dead-letter queue.

[Cancel](#) [Create subscription](#)

We can n number of subscription for multiple email address to send notification.

Once created, we must **confirm** the subscription in our email. Until, it shows **pending confirmation**.

MytopicforEC2

EditDeletePublish message

Details

Name
MytopicforEC2

ARN
arn:aws:sns:ap-south-1:314383175192:MytopicforEC2

Display name
MytopicforEC2

Type
Standard

Topic owner
314383175192

Subscriptions

Access policyData protection policyDelivery policy (HTTP/S)Delivery status loggingEncryptionTagsIntegrations

Subscriptions (1)

EditDeleteRequest confirmationConfirm subscriptionCreate subscription

ID	Endpoint	Status	Protocol
Pending confirmation	nithyaaisubs@gmail.com	Pending confirmation	EMAIL

Open the **respective email address** provided for subscription → open the **email from SNS** → Click **Confirm Subscription**.

AWS Notification - Subscription ConfirmationInbox x

mytopic

You have chosen to subscribe to the topic: arn:aws:sns:us-east-1:969578073036:mytopic To confirm this subscription, click or visit the link below (If this was i

MytopicforEC2 <no-reply@sns.amazonaws.com>

to me

You have chosen to subscribe to the topic:
arn:aws:sns:ap-south-1:314383175192:MytopicforEC2

...

To confirm this subscription, click or visit the link below (If this was in error no action is necessary):
[Confirm subscription](#)

Please do not reply directly to this email. If you wish to remove yourself from receiving all future SNS subscription confirmation requests please send an email to [sns-opt-out](#)

Simple Notification Service

Subscription confirmed!

You have successfully subscribed.

Your subscription's id is:

arn:aws:sns:ap-south-1:314383175192:MytopicforEC2:a234cbcc-3a17-4736-af0b-5e5d1bdb3fbf

If it was not your intention to subscribe, [click here to unsubscribe](#).

Once confirmed → Status become **confirmed**.

MytopicforEC2

[Edit](#) [Delete](#) [Publish message](#)

Details

Name
MytopicforEC2

ARN
arn:aws:sns:ap-south-1:314383175192:MytopicforEC2

Display name
MytopicforEC2

Type
Standard

Topic owner
314383175192

Subscriptions

[Access policy](#)

[Data protection policy](#)

[Delivery policy \(HTTP/S\)](#)

[Delivery status logging](#)

[Encryption](#)

[Tags](#)

[Integrations](#)

Subscriptions (1)

Search

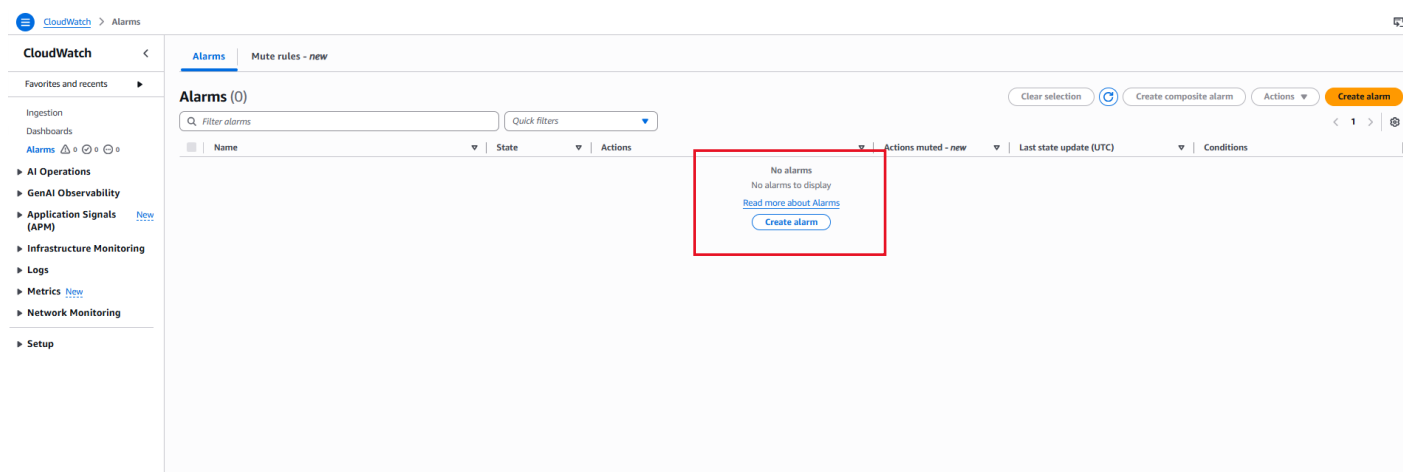
[Edit](#) [Delete](#) [Request confirmation](#) [Confirm subscription](#) [Create subscription](#)

ID	Endpoint	Status	Protocol
a234cbcc-3a17-4736-af0b-5e5d1bdb3fbf	nithyaaisubs@gmail.com	Confirmed	EMAIL

Step no:4 In the console → Search **CloudWatch** → click it.



Go to **alarms** → Create alarm.



Go to the EC2 that we have created for monitoring using cloudwatch and copy its **instance id**.

Instances (1/1) [Info](#)

Saved filter sets

Choose filter set ▼

Find Instance by attribute or tag (case-sensitive)

☒	Name 🔗	Instance ID	Instance state 🔍	Instance type	Status check	Alarms
☒	My-cloudwatch-instance	i-0be319a6930ca2e09	Running 🔍	t3.micro	3/3 checks passed 🔍	View

i-0be319a6930ca2e09 (My-cloudwatch-instance)

[Details](#) | [Status and alarms](#) | [Monitoring](#) | [Security](#) | [Networking](#) | [Storage](#) | [Tags](#)

▼ Instance summary [Info](#)

Instance ID

[🔗](#) i-0be319a6930ca2e09

Public IPv4 address

[🔗](#) 13.234.225.146 | [open address](#) [🔗](#)

Now comeback to alarms → **Select metric** → **Paste the instance id** → Enter → Choose: **EC2> Per-instance Metrics**.

Specify metric and conditions

Metric

Graph

Preview of the metric or metric expression and the alarm threshold.

[Select metric](#)

Search for any metric, dimension, resource id or account id

[🔗](#) i-0be319a6930ca2e09 [🔗](#) [✕](#)

EBS > Per-Volume Metrics with Instance ID

7

EC2 > Per-Instance Metrics

22

Choose: **CPUUtilization**

☐	My-cloudwatch-instance	i-0be319a6930ca...	StatusCheckFailed_AttachedEBS 🔍	No alarms
☐	My-cloudwatch-instance	i-0be319a6930ca...	MetadataNoToken 🔍	No alarms
☒	My-cloudwatch-instance	i-0be319a6930ca...	CPUUtilization 🔍	No alarms

Provide,

Time period: **one minute**

Threshold type: **Static**

Notify when the CPU Utilization is **greater**.

Threshold: **2%** (normally the threshold value is 80%, to check it is working we provide 2%).

Next.

Specify metric and conditions

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.

Percent

2%

1.07%

0.137%

09:30 09:45 10:00 10:15 10:30 10:45 11:00 11:15 11:30 11:45 12:00 12:15

CPUUtilization

Namespace

AWS/EC2

Metric name

CPUUtilization

Instanceld

i-0be319a6930ca2e09

Instance name

My-cloudwatch-instance

Statistic

Average

Period

1 minute

Conditions

Threshold type

☒ Static

Use a value as a threshold

☐ Anomaly detection

Use a band as a threshold

Whenever CPUUtilization is...

Define the alarm condition.

☒ Greater

> threshold

☐ Greater/Equal

>= threshold

☐ Lower/Equal

<= threshold

☐ Lower

< threshold

than...

Define the threshold value.

2

Must be a number.

Additional configuration

Cancel

Next

Configure,

Alarm state: **In alarm**

(The metric or expression is outside the defined Threshold)

Send notification topic: **Existing SNS Topic → MytopicforEC2**

Next

Configure actions

Notification

Alarm state trigger

Define one alarm state that will trigger this action.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

☒ **In alarm**
The metric or expression is outside of the defined threshold.

☐ **OK**
The metric or expression is within the defined threshold.

☐ **Insufficient data**
The alarm has just started or not enough data is available.

Send a notification to the following SNS topic

Send a notification to the following SNS topic
 Define the SNS (Simple Notification Service) topic that will receive the notification.

☒ **Select an existing SNS topic**

☐ **Create a new SNS topic**

☒ Select an existing SNS topic

☐ Create new topic

☐ Use topic ARN to notify other accounts

☐ Create new topic

☐ Use topic ARN to notify other accounts

Send a notification to...

MytopicforEC2

Send a notification to...

Q

MytopicforEC2

X

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Only topics belonging to this account are listed here. All persons and applications subscribed to the selected topic will receive notifications.

Email (endpoints)
nithyaaisubs@gmail.com - [View in SNS Console](#)

[Add notification](#)

[Email \(enquiries\)](#)
nithyaaisubs@gmail.com - [View in SNS Console](#)

Lambda action

[Add Lambda action](#)

Auto Scaling action

[Add Auto Scaling action](#)

EC2 action

EC2 action

[Add EC2 action](#)

Systems Manager action [Learn more](#)

Investigation action - **new** [Info](#)

This action will create an investigation when the alarm is **In alarm** state.

[Add investigation action](#)

This action will create an investigation when the alarm is **In alarm** state.

[Add investigation action](#)

[Add investigation action](#)

[Cancel](#)
[Previous](#)
[Next](#)

Cancel Previous Next

Provide a name for the alarm: **High CPU utilization.**

Add alarm details

Name and description	
Alarm name	

Alarm name

High CPU utilization

Alarm description - optional [View formatting guidelines](#)

High CPU utilization

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

Alarm description - optional [View formatting guidelines](#)

Edit

Preview

This is to inform the CPU utilization has been reached the threshold.

[Edit](#) [Preview](#)

This is to inform the CPU utilization has been reached the threshold.

This is to inform the CPU utilization has been reached the threshold.

Up to 1024 characters (69/1024).

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Up to 1024 characters (69/1024).

 Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

① Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

Tags - optional [Info](#)

No tags associated with the resource.

No tags associated with the resource.

[Add new tag](#)

You can add up to 50 tags.

You can add up to 50 tags. Cancel Previous Next

[Cancel](#)
[Previous](#)
[Next](#)

Review and Create Alarm.

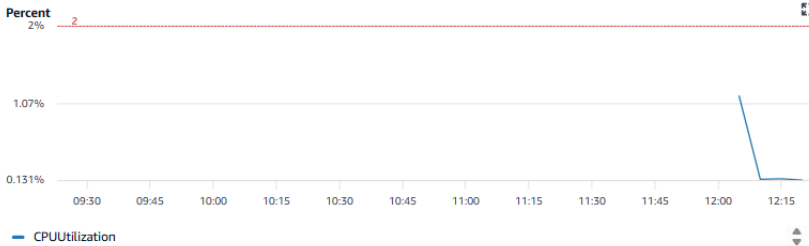
Preview and create

Step 1: Specify metric and conditions

Metric

Graph

This alarm will trigger when the blue line goes above the red line for 1 datapoints within 1 minute.



Namespace

AWS/EC2

Metric name

CPUUtilization

InstanceId

i-0be319a6930ca2e09

Instance name

My-cloudwatch-instance

Statistic

Average

Period

1 minute

Conditions

Threshold type

Static

Whenever CPUUtilization is
Greater (>)

than...

2

► Additional configuration

Step 2: Configure actions

[Edit](#)

Actions

Notification

When in alarm, send a notification to "MytopicforEC2"

Step 3: Add alarm details

[Edit](#)

Alarm details

Name

High CPU utilization

Description

This is to inform the CPU utilization has been reached the threshold.

► Tags (0)

Markdown formatting is only applied when viewing your alarm in the console. The description will remain in plain text in the alarm notifications.

[Cancel](#)[Previous](#)[Create alarm](#)

Once the alarm is created → the **status** will be **insufficient** (It means that the alarm has just started, does not have sufficient data).

Alarms (1)

[Clear selection](#)[Create composite alarm](#)[Actions](#)[Create alarm](#)[Quick filters](#)

☐	Name	State	Actions	Actions muted - new	Last state update (UTC)	Conditions
☐	High CPU utilization	Insufficient data	Actions enabled	-	2026-04-16 12:24:05	CPUUtilization > 2 for 1 datapoints within 1 minute

Step no:5 Now go to our instance → connect → Now we are going to make the CPU Utilization more than 2%.

To create a fake stress on EC2,

Switch user: **sudo su**

Install Stress package: **yum install stress -y**

To create a fake stress on CPU: **stress -cpu 1 -timeout 30s**

```
ec2-user@ip-172-31-34-17 ~]$ sudo su
root@ip-172-31-34-17 ec2-user[# yum install stress -y
Amazon Linux 2023 Kernel Livepatch repository
Dependencies resolved.

Package Architecture
-----
Installing:
stress x86_64

Transaction Summary
-----
Install 1 Package

Total download size: 34 k
Installed size: 68 k
Downloading Packages:
stress-1.0.7-2.amzn2023.0.1.x86_64.rpm
-----
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
  Preparing      : 
  Installing     : stress-1.0.7-2.amzn2023.0.1.x86_64
  Running scriptlet: stress-1.0.7-2.amzn2023.0.1.x86_64
  Verifying      : stress-1.0.7-2.amzn2023.0.1.x86_64

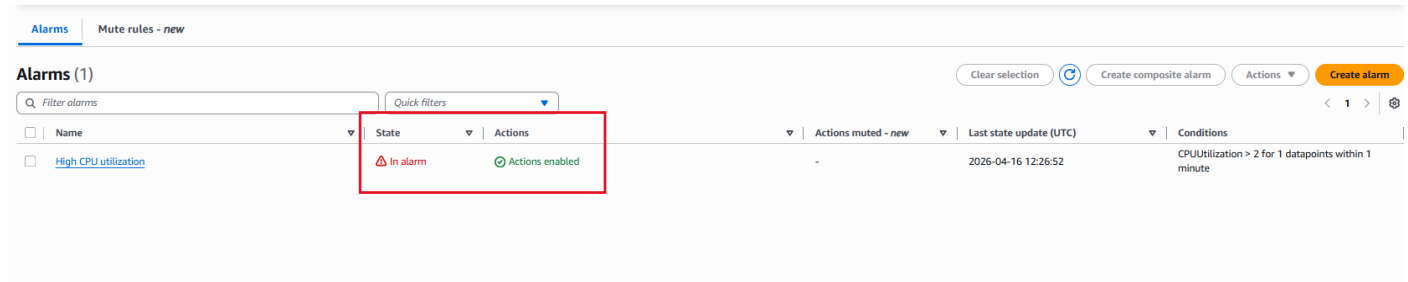
Installed:
  stress-1.0.7-2.amzn2023.0.1.x86_64

Complete!
root@ip-172-31-34-17 ec2-user[# stress --cpu 1 --timeout 30s
stress: info: [26296] dispatching hogs: 1 cpu, 0 io, 0 vm, 0 hdd
stress: info: [26296] successful run completed in 30s
root@ip-172-31-34-17 ec2-user[#
```

i-0be319a6930ca2e09 (My-cloudwatch-instance)

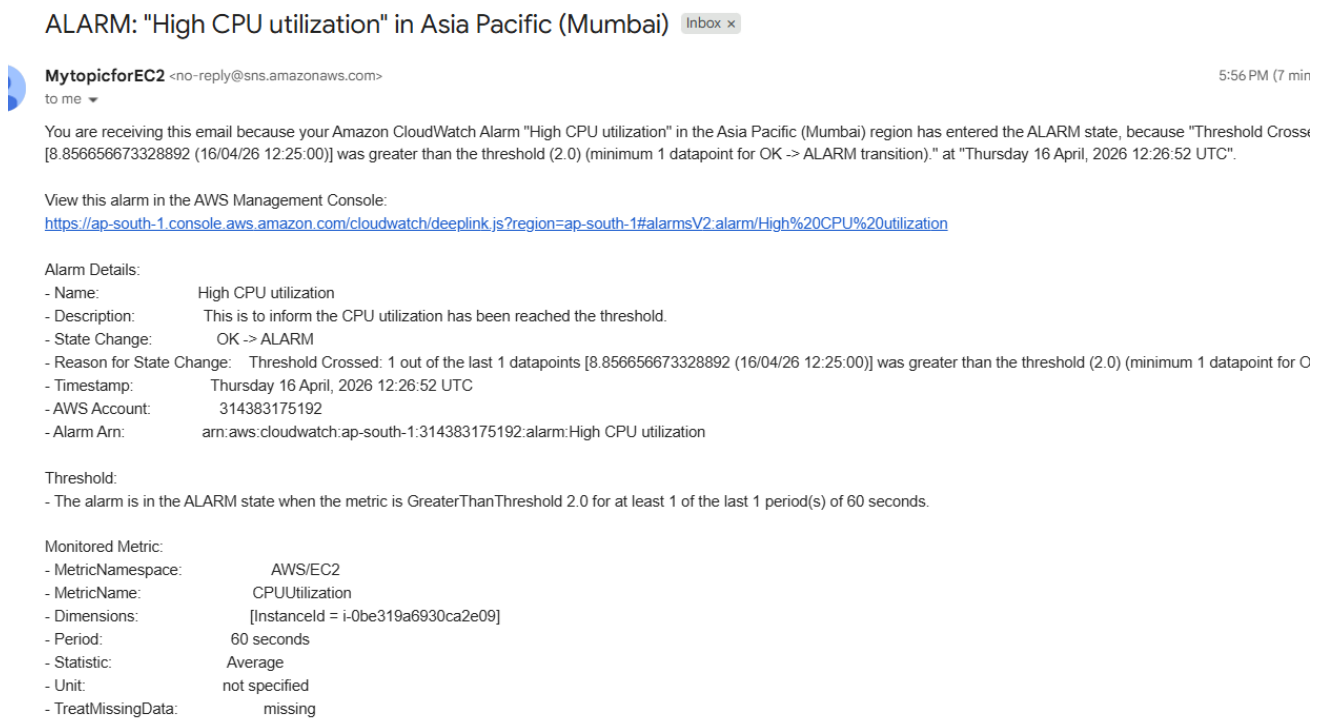
PublicIPs: 13.234.225.146 PrivateIPs: 172.31.34.17

Once the stress has been implemented , the status becomes **in alarm** (as the threshold value has been crossed).



The screenshot shows the AWS Alarms console. At the top, there's a tab for 'Alarms' and a link for 'Mute rules - new'. Below this, the 'Alarms (1)' section is visible. A search bar and a 'Quick filters' dropdown are present. A table lists the alarm 'High CPU utilization'. The 'State' column shows 'In alarm' with a red triangle icon, and the 'Actions' column shows 'Actions enabled' with a green checkmark icon. Other columns include 'Actions muted - new', 'Last state update (UTC)' (2026-04-16 12:26:52), and 'Conditions' (CPUUtilization > 2 for 1 datapoints within 1 minute).

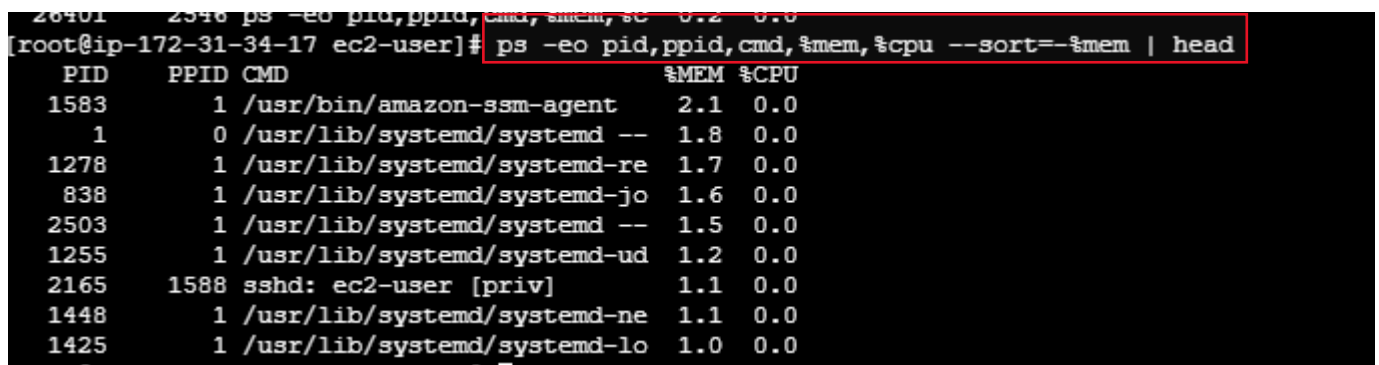
We will receive an email notification regarding the **CPU Utilization**.



The screenshot shows an email notification from 'MytopicforEC2' to 'me'. The subject is 'ALARM: "High CPU utilization" in Asia Pacific (Mumbai)'. The body text states: 'You are receiving this email because your Amazon CloudWatch Alarm "High CPU utilization" in the Asia Pacific (Mumbai) region has entered the ALARM state, because "Threshold Crossed: 1 out of the last 1 datapoints [8.856656673328892 (16/04/26 12:25:00)] was greater than the threshold (2.0) (minimum 1 datapoint for OK -> ALARM transition)." at "Thursday 16 April, 2026 12:26:52 UTC"'. A link is provided to view the alarm in the AWS Management Console. Below this, 'Alarm Details' are listed: Name: High CPU utilization, Description: This is to inform the CPU utilization has been reached the threshold, State Change: OK -> ALARM, Reason for State Change: Threshold Crossed: 1 out of the last 1 datapoints [8.856656673328892 (16/04/26 12:25:00)] was greater than the threshold (2.0) (minimum 1 datapoint for O, Timestamp: Thursday 16 April, 2026 12:26:52 UTC, AWS Account: 314383175192, Alarm Arn: arn:aws:cloudwatch:ap-south-1:314383175192:alarm:High CPU utilization. The 'Threshold' section states: 'The alarm is in the ALARM state when the metric is GreaterThanThreshold 2.0 for at least 1 of the last 1 period(s) of 60 seconds.' The 'Monitored Metric' section lists: MetricNamespace: AWS/EC2, MetricName: CPUUtilization, Dimensions: [InstanceId = i-0be319a6930ca2e09], Period: 60 seconds, Statistic: Average, Unit: not specified, TreatMissingData: missing.

Use the commands: **ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | head**

We can see the memory utilization of top 10 in descending order.



```
26401 2348 ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | head
[root@ip-172-31-34-17 ec2-user]# ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%mem | head
  PID   PPID  CMD                                %MEM  %CPU
  1583     1 /usr/bin/amazon-ssm-agent           2.1    0.0
    1     0 /usr/lib/systemd/systemd --         1.8    0.0
  1278     1 /usr/lib/systemd/systemd-re         1.7    0.0
   838     1 /usr/lib/systemd/systemd-jo         1.6    0.0
  2503     1 /usr/lib/systemd/systemd --         1.5    0.0
  1255     1 /usr/lib/systemd/systemd-ud         1.2    0.0
  2165   1588 sshd: ec2-user [priv]                1.1    0.0
  1448     1 /usr/lib/systemd/systemd-ne         1.1    0.0
  1425     1 /usr/lib/systemd/systemd-lo         1.0    0.0
```

Use the command: `ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu | head`

This shows the CPU utilization in descending order (-).

```
psstress: info: [26296] successful run completed in 30s
[root@ip-172-31-34-17 ec2-user]# ps -eo pid,ppid,cmd,%mem,%cpu --sort=-%cpu
  PID     PPID  CMD                                %MEM %CPU
    1         0 /usr/lib/systemd/systemd --      1.8  0.0
    2         0 [kthreadd]                        0.0  0.0
    3         2 [rcu_gp]                          0.0  0.0
    4         2 [rcu_par_gp]                     0.0  0.0
    5         2 [slub_flushwq]                   0.0  0.0
    6         2 [netns]                          0.0  0.0
    8         2 [kworker/0:0H-events_highpri]  0.0  0.0
   10         2 [mm_percpu_wq]                   0.0  0.0
   11         2 [rcu_tasks_kthread]             0.0  0.0
   12         2 [rcu_tasks_rude_kthread]        0.0  0.0
   13         2 [rcu_tasks_trace_kthread]       0.0  0.0
   14         2 [ksoftirqd/0]                   0.0  0.0
   15         2 [rcu_preempt]                   0.0  0.0
```

Once the system is back to normal or less than the threshold, the status become **OK**.

Alarms (1)

Filter alarms		Quick filters	
☐ Name	State	Actions	Action
☐ High CPU utilization	✔ OK	✔ Actions enabled	-